HeartSafe Agent: An Equity-Aware AI Agent for Trustworthy Remote Cardiac Monitoring and Closed-Loop Care in Rural Canada

Team #40

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Hackathon Theme: AI for Good in the Era of Agents
Theme Area: Theme 4 – AI for Social Impact: Health, Agriculture & Inclusive Innovation
Challenge: Challenge 1 – AI for Better & More Equitable Healthcare

1. Problem Being Addressed

Heart disease is the second leading cause of death in Canada, claiming 57,890 lives in 2023. Rural and remote communities, including many First Nations, Métis and Inuit communities, face added barriers to specialized cardiac care, diagnostic imaging and timely follow-up. Nationally, Canadians already wait a median of 59 days for an MRI. First Nations people have about 2.5 times the cardiovascular disease prevalence of non-First Nations people, and in Ontario, rural patients with stable ischemic heart disease had 29% fewer cardiology visits than urban patients in the year after angiography.

Remote monitoring platforms help close this gap. Medly, from the University Health Network, already supports heart-failure patients in First Nations communities in Northern Ontario: after discharge, they send daily weight, blood pressure, heart rate and symptoms from home and receive alerts. AI-powered wearable ECG and medical image analysis could add further insight. But home readings are often poorly captured (a new scale, weight typed in pounds, a misplaced cuff, missed days), and AI models are usually validated on limited populations. Both can trigger false alarms or fail silently, and the communities with the fewest specialists are the least able to catch these failures.

Detecting an abnormality also does not guarantee that the patient receives care. In resource-constrained settings, uncertain results, delayed reviews and unresolved referrals can quietly fall through the cracks, and a patient may never receive or act on the clinician's instruction, widening existing health inequities.

2. Proposed AI Solution and Approach

HeartSafe Agent is an equity-aware agentic AI layer that sits on top of existing remote monitoring platforms and connects their alerts and cardiac AI predictions to trustworthy, accountable follow-up (Figure 1). It does not replace the platform or the clinical team. It combines three inputs: the platform's daily readings and alerts; a wearable single-lead ECG model that detects concerning patterns at home; and a medical image segmentation model that quantifies cardiac structure and function when imaging is performed, typically at a regional centre after referral.

Target population. Adults in rural and remote communities, discharged within the last 14 days after more than 48 hours in hospital for decompensated heart failure, NYHA class II–IV, and able to take readings alone or with a caregiver, monitored for 30–90 days. Exclusions follow a published Medly trial: dementia, long-term care, life expectancy under one year, transplant listing or ventricular assist device, and dialysis. NYHA class is used because it is assessed at the bedside, without an echocardiogram.

Around these inputs, the agent plans and carries out four tasks:

- **Evaluates reliability.** For every alert or result, it checks data quality (implausible values, unit and cuff errors, device changes), signal and image quality, prediction uncertainty, device or scanner shift, and physiological plausibility. It then decides whether to accept the result, request a repeat, or escalate. A reliable ECG is also named: new atrial fibrillation is reported to the care team, atrial fibrillation already in the enrollment record is not, and the heart rate is checked against the patient's own limits. A suspected new scale is confirmed by a nurse before baselines change, and only a clinician can close a case.
- **Coordinates escalation.** It prepares structured review requests for clinicians, including the key findings, the reason for escalation, the patient's context (NYHA class, medications, caregiver), reliability caveats, plain-language ECG and MRI report summaries and visual explanations where available (e.g., uncertainty maps). Chest pain or fainting with a reliable abnormal ECG is an emergency: the patient is told to call 911 if they feel worse, and a clinician must start a video call within 15 minutes.
- **Closes the loop.** It follows each escalation from creation through clinician acknowledgement, review and resolution, including when a patient is referred to a regional centre and the result must return home. It then confirms that

the patient received the instruction, acted on it, and that subsequent readings improved, contacting a caregiver or community health worker when the patient does not respond. A follow-up watch re-checks every case every two hours, identifies stalled cases and sends reminders according to clinician-approved protocols. Patients can also log their sleep and the breathing exercises in their care team's plan; sleep worse than their own usual opens a routine nurse check-in call, never a clinical alert.

- **Investigates equity.** It continuously compares model performance, escalation outcomes and time to close the loop across subgroups (e.g., remoteness, sex, age, device, scanner vendor), using confidence intervals. It explores likely causes of any gaps, such as data under-representation, quality differences or shifts in model attention, and recommends safeguards for human approval. Once an administrator approves a safeguard, patients in the affected subgroup receive more conservative escalation and closer follow-up.

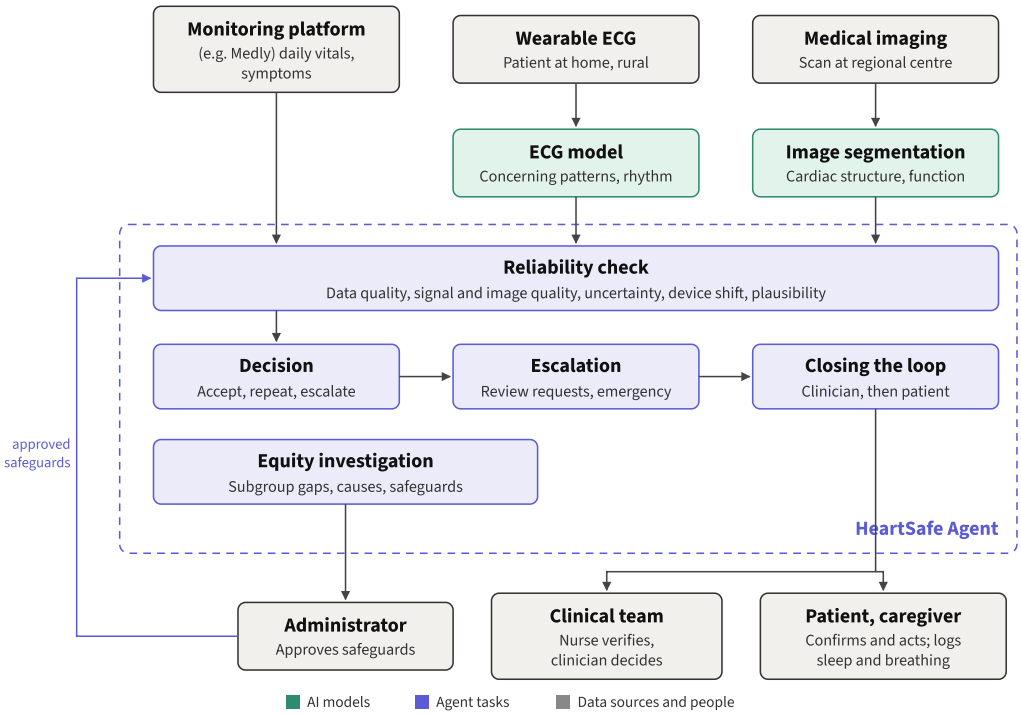


Figure 1: HeartSafe Agent workflow. Platform alerts, wearable ECG (home) and medical imaging (regional centre) feed the agent. It checks each result's reliability, decides whether to accept, repeat or escalate, and tracks escalations until a clinician resolves them and the patient has acted. Its equity investigation proposes safeguards for administrator approval.

Implementation and evaluation. The prototype combines a rule-based agent (reliability checks, case tracker, patient loop, follow-up watch and equity monitor) with our existing nnU-Net medical image segmentation pipeline, a single-lead ECG classifier and a rhythm model (normal, atrial fibrillation, other). Alert and safety rules were reviewed by a practising cardiovascular anesthesiologist and a practising cardiologist, who added diastolic blood pressure, oxygen saturation, further symptoms (orthopnea, paroxysmal nocturnal dyspnea, chest pain, syncope), diuretic adherence and post-discharge potassium and creatinine, required nurse verification of device changes, and defined the emergency level. Reviews are written from templates; an optional LLM can reword them using read-only tools, but it cannot change thresholds, severities or actions. We validate the models on public, de-identified datasets with demographic and device metadata: PTB-XL for clinical 12-lead ECG subgroup analysis, patient-recorded single-lead handheld ECGs from the PhysioNet 2017 Challenge to test device shift, and multi-vendor, multi-centre cardiac imaging data (M&Ms). Because platform data are not public, a simulated cohort of 2,000 post-discharge patients over 90 days, modelled on Medly's published inputs, screened with our inclusion criteria and seeded with device and entry errors, demonstrates the rural workflow. We evaluated, against the monitoring platform alone:

- detection of unreliable predictions and of alerts caused by faulty readings: 86% of alerts caused by faulty readings were stopped before reaching a nurse, and the reliability checks caught every injected ECG fault and 92% of injected MRI faults;
- appropriateness of escalation decisions: every true heart-failure episode was escalated, with 60 instead of 75 false alarms per 100 patient-months; 94% of new atrial fibrillation episodes were reported, a median of 2 days after onset;

- detection of subgroup performance gaps: slower acknowledgement and more unusable ECGs in remote communities, and lower MRI segmentation accuracy on an unseen scanner vendor and in patients aged 60 and over;
- identification of unresolved cases: 534 of 541 stalled cases were flagged by a reminder, against none without tracking;
- simulated follow-up completion and patient loop closure: 96% of instructions were completed, against 76% with a single message;
- simulated time to acknowledgement and to case resolution: a median of 6.7 instead of 12.8 hours to acknowledge a true episode, and 97% instead of 85% of cases resolved within their deadline;
- extra workload from tighter thresholds and nurse verification: 100 nurse items per 100 patient-months instead of 107, including 1.6 scale verifications and 5.3 sleep check-in calls.

3. Connection to the Hackathon Theme and Canadian Community Impact

HeartSafe Agent addresses Theme 4, Challenge 1, by applying agentic AI to more equitable cardiac care. Prior work has tackled parts of this problem: quality-control agents for image segmentation (e.g., ARGUS), subgroup-aware human–AI decision allocation (e.g., PecMan), and commercial monitoring of deployed imaging AI (e.g., Lattice Health, Aival Monitor). HeartSafe Agent unifies trust assessment, equity investigation and closed-loop follow-up across home monitoring, wearable and imaging AI, along a single rural care pathway. Because it works on top of platforms already used in Canada, such as Medly, TELUS Home Health Monitoring and Cloud DX, it can be added without replacing existing care.

For rural and remote Canadian communities, this means:

- concerning findings are less likely to be overlooked, and false alarms from faulty readings are caught;
- scarce specialist and nursing time is directed to the cases that need it;
- every escalation ends with a confirmed action by the patient, not only a clinician's note;
- AI performance gaps affecting specific groups are surfaced rather than hidden.

Responsible AI. The agent never diagnoses, and qualified clinicians make all clinical decisions; a named ECG rhythm is reported for confirmation on a 12-lead ECG. Alert rules are reviewed by clinicians, every agent action is logged for audit, subgroup performance is reported transparently, and only de-identified public and simulated data are used. Public datasets do not record rural or Indigenous status, so extending the equity audit to these groups is planned future work. That work would be done with community partners under Indigenous data-governance principles (OCAP®).

Scalability. New platforms, sites, devices or cardiac models can be added through a platform adapter and local escalation protocols, while the reliability, follow-up and equity layers stay the same. Beyond Canada, heart disease is the leading cause of death in Mexico (192,518 deaths in 2024) and in the United States, and rural communities across North America can face similar gaps in cardiac follow-up. Where home readings are often poorly captured, the data-quality checks matter even more; adapting HeartSafe to these health systems, their connectivity and their privacy rules is future work.

HeartSafe Agent shows how AI agents can move beyond prediction to accountable action, helping close the gap between AI innovation and equitable cardiac care in Canada.

Statistics: Statistics Canada (2026), heart disease deaths in 2023; Canadian Institute for Health Information (2025), median MRI wait times; Anand et al., The Lancet (2001), as cited in Bresee et al., CMAJ (2014), First Nations cardiovascular disease prevalence; Tran et al., Circulation: Cardiovascular Quality and Outcomes (2014), rural and urban cardiology visits. INEGI (2025), registered deaths in Mexico in 2024; CDC National Center for Health Statistics, leading causes of death in the United States.

Platform and population: University of Toronto Institute for Social Innovation (2024), Medly with the Weeneebayko Area Health Authority; Sioux Lookout Meno Ya Win Health Centre and Sioux Lookout First Nations Health Authority, regional heart-failure care with Medly; Medly post-discharge trial, ClinicalTrials.gov NCT03358303 (inclusion and exclusion criteria).

Datasets and methods: PTB-XL (Wagner et al., Scientific Data, 2020); PhysioNet 2017 Challenge (Clifford et al., Computing in Cardiology, 2017); M&Ms (Campello et al., IEEE Transactions on Medical Imaging, 2021); nnU-Net (Isensee et al., Nature Methods, 2021).